PRODUCER GUIDE

RatioAvis 1.1.1

The *cheapest* diet for your broilers, step by step

A practical guide to formulating, mixing and costing your farm's feed with RatioAvis, explained through one real case from start to finish.

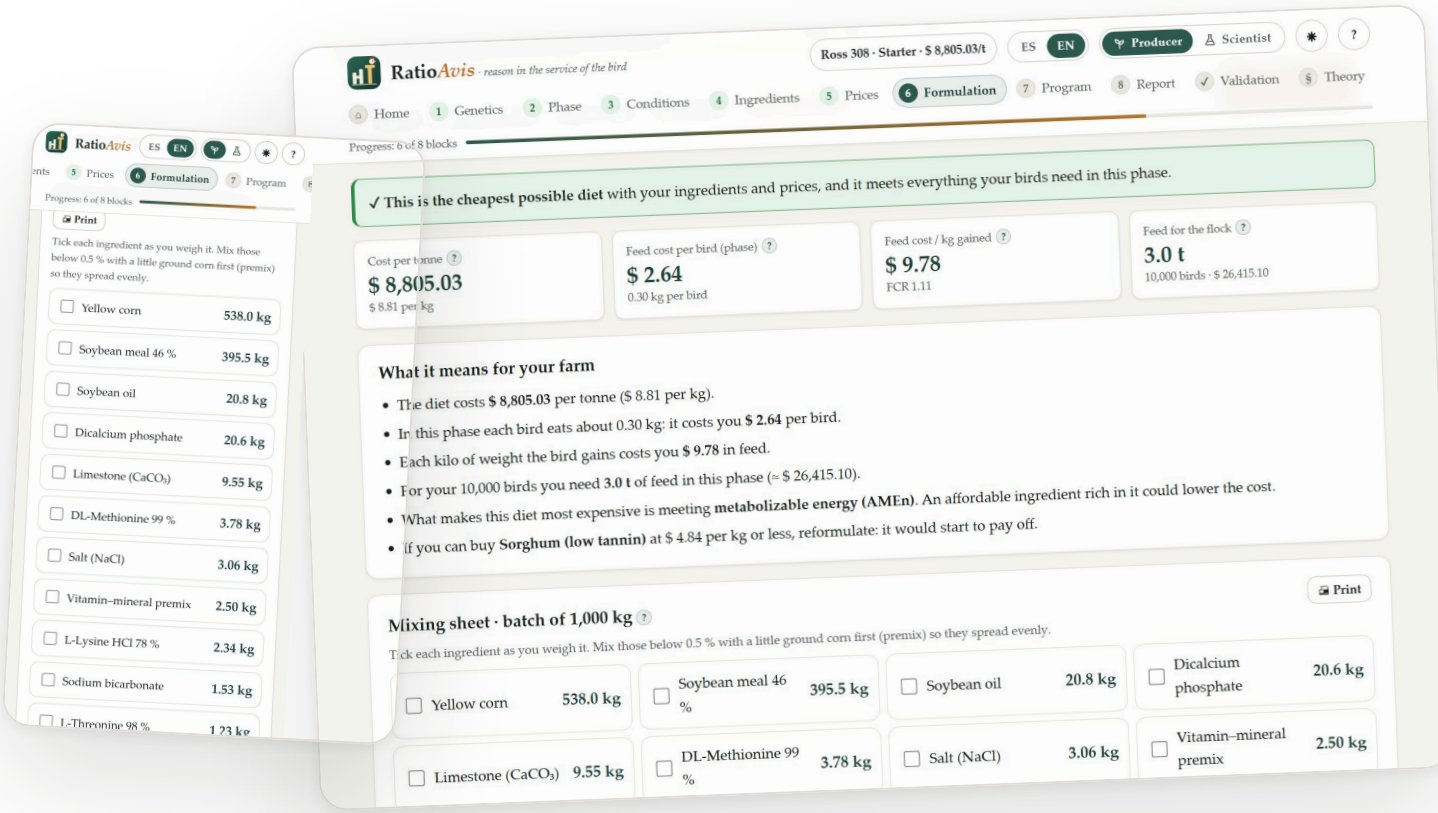
Open the app: **hectortecumshe-ai.github.io/RatioAvis**

1st edition · October 2026 · covers RatioAvis 1.1.1

DOI 10.5281/zenodo.23123050 · MIT License

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There is a companion guide for research and teaching: the Scientist Guide.



1	2	3	4	5	6	7	8
Genetics	Phase	Conditions	Ingredients	Prices	Formulation	Program	Report

BEFORE READING

How to use this guide

Who it is for, how it is organised and what each sign on the page means.

This guide is for whoever decides what the birds on a farm eat: the producer, the feed-mill manager, the advisor who works with several farms. You need no coding or maths; you need to know your farm.

It follows the app's order: one chapter for each block in the top bar, 1 to 8, each with its own colour. Every example uses **a single case**, a high-altitude farm with 10,000 broilers, so the numbers in one chapter are the ones that feed the next. Read it start to finish and you will have formulated, mixed and costed a real diet.

Conventions

WRITTEN AS	MEANS
⚙ Formulate	An app button or field, with its text exactly as on screen.
Report › Save project	A path: open the first item, then the next.
2200	A value you type as shown.
1	A marker on a screenshot; it is explained below with the same number.

The five boxes

◆ **ON THE FARM**

The worked case, step by step, with the real numbers the app returns.

✓ **TIP**

Shortcuts and good practice that save time or money.

! **CAUTION**

What, if overlooked, can lead to a wrong diet.

? **WHY?**

The reason behind a recommendation, in two or three sentences and without formulas.

→ **DECISION RULE**

If you see this on screen… **then** do this. They are guides: what you know about your farm always comes first.

How to cite

The software: Mojica-Zárate, H. T., Barrera-Guzmán, L. Á., & Velázquez-Vázquez, G. (2026). *RatioAvis: Guided least-cost diet formulation for broiler chickens* (Version 1.1.1) [Software]. Zenodo. https://doi.org/10.5281/zenodo.23123050

This guide: Mojica-Zárate, H. T., Barrera-Guzmán, L. Á., & Velázquez-Vázquez, G. (2026). *RatioAvis: Producer Guide* (1st ed.). Distributed with RatioAvis 1.1.1. https://hectortecumshe-ai.github.io/RatioAvis/instructivo-productor.html

🔒 **YOUR DATA NEVER LEAVE YOUR COMPUTER**

RatioAvis does every calculation inside the browser. Your prices, formulas and farm name are not sent to any server; they are stored only on your device.

BEFORE THE BLOCKS

0 Getting started

What RatioAvis does, what data to have at hand and how the screen is laid out.

0.1 What RatioAvis does

You tell it which birds you have, their growth phase, which ingredients you can buy and at what price. RatioAvis tries every possible combination and returns **the cheapest mix that covers everything your birds need**, with the exact kilos to weigh for your mixer and what each bird costs you.

> ? **WHY IS THE CHEAPEST NOT THE POOREST?**
>
> Because the low price is sought only *among formulas that already meet* the energy, protein, amino acids and minerals required by your genetic line's table. No formula that falls short on a nutrient can win, however cheap.

0.2 What to have at hand

DATA	WHERE TO FIND IT	BLOCK
Genetic line of your chicks	Hatchery invoice or chick box	1
Feed intake and weight gain per phase	Your records from past flocks (or the table values)	2
Altitude and house temperature	Phone GPS · house thermometer	3
Number of birds and mixer capacity	Your flock · mixer nameplate	3
Ingredients you can buy and their delivered price	Your suppliers (freight included)	4 · 5

0.3 Opening the app

Open hectortecumshe-ai.github.io/RatioAvis in any modern browser (Chrome, Edge, Firefox or Safari) on a computer, tablet or phone. There is nothing to install and no account to create. Once open, it keeps working even if the internet drops.

0.4 A tour of the screen

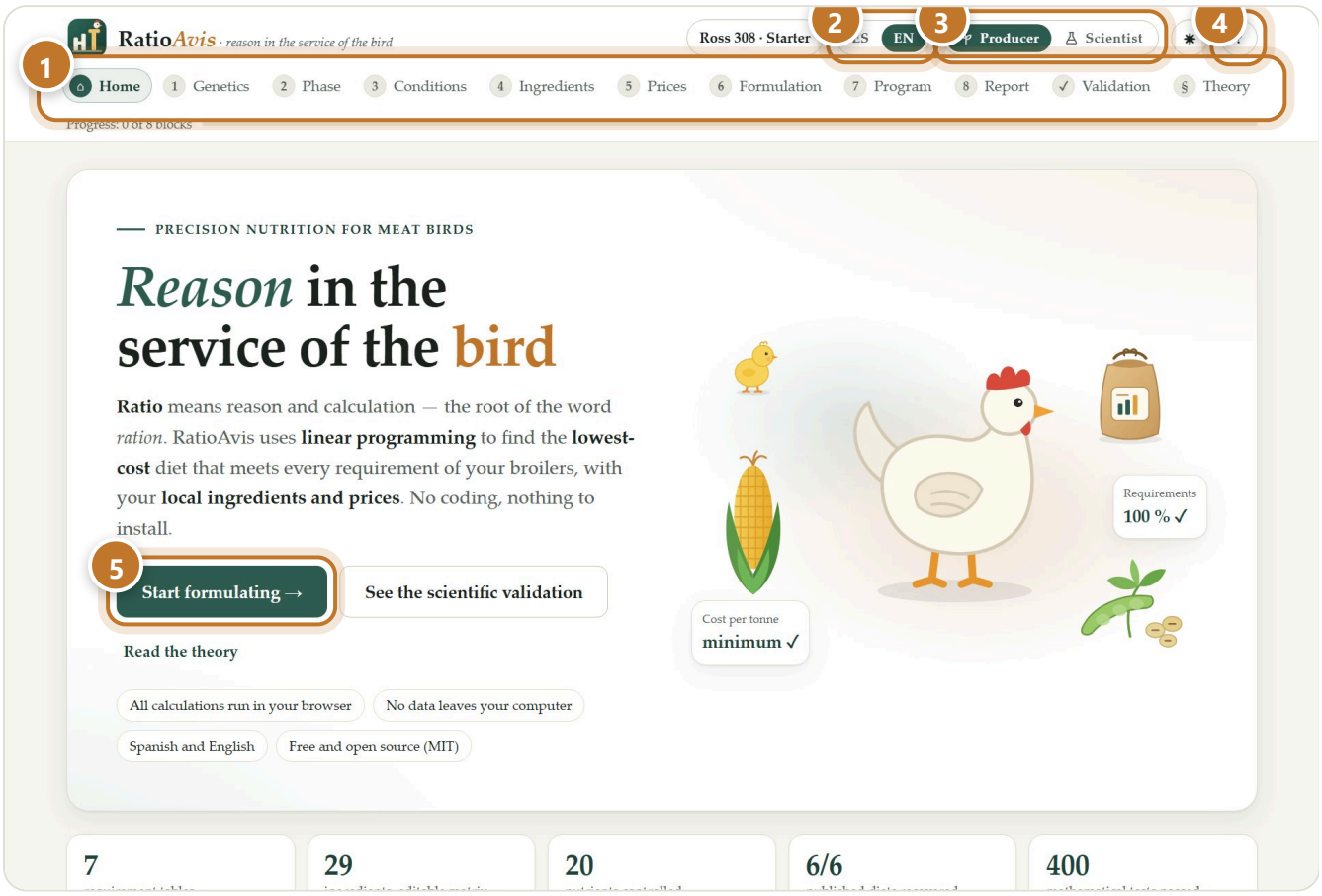


FIGURE 0.1 **The home screen.** The top bar stays with you in every block.

1. **Block bar.** The eight steps in order. The current block is highlighted and those already visited turn green.
2. **ES / EN.** Switch language at any time; your data are kept.
3. **Producer / Scientist.** This guide uses the **Producer** view: the essentials, in plain language.
4. **Help (?).** A summary of how to use the app.
5. **Start formulating.** Takes you to Block 1.

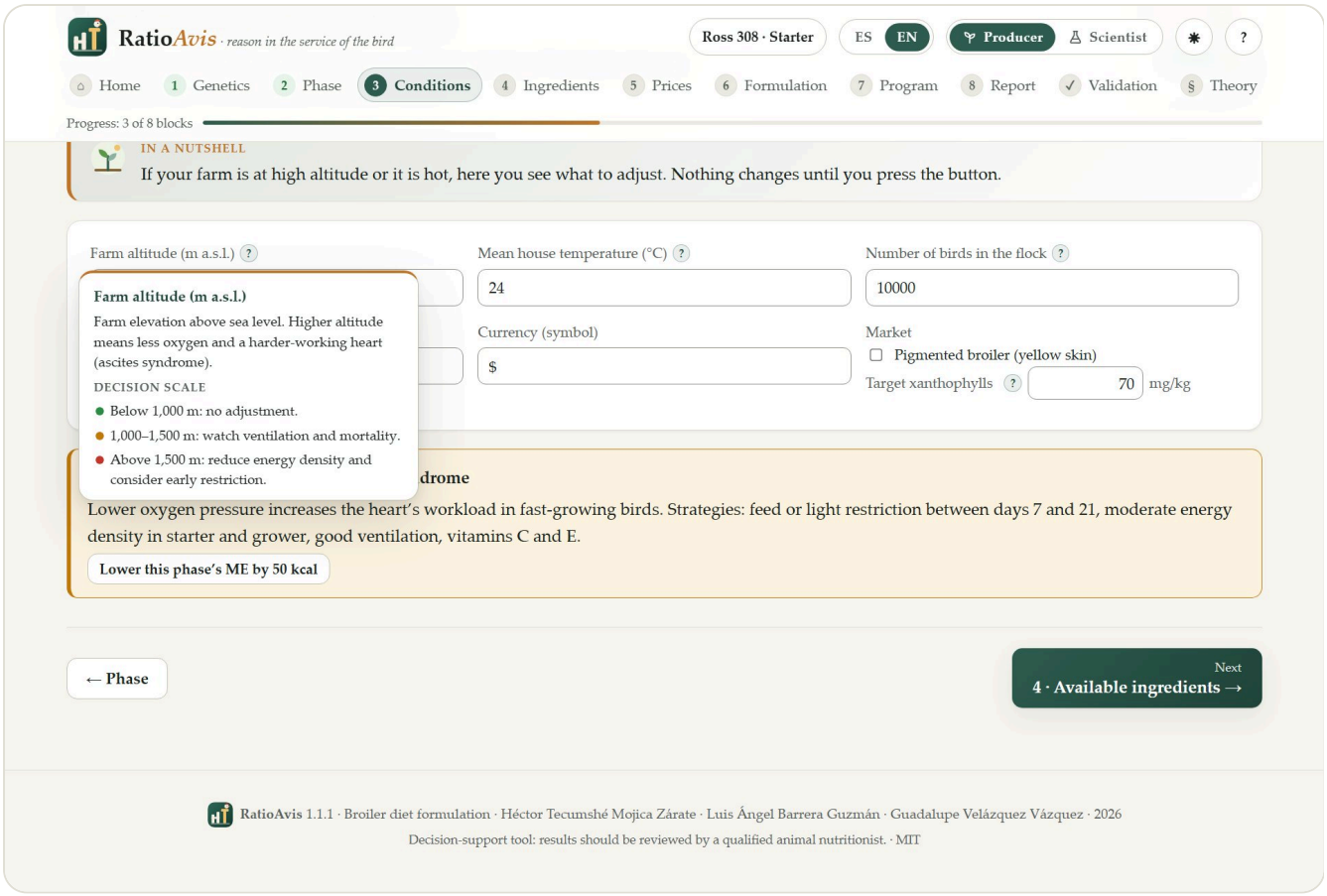


FIGURE 0.2 **The ? sign next to each field.** Hovering (or tapping it on a phone) explains what it is and shows a colour-coded **decision scale**: green, amber and red.

0.5 The case in this guide

So that every step has concrete numbers, we will follow one farm from start to finish:

LINE **Ross 308**	SEX **As-hatched**	PHASE **Starter · 0–10 d**
ALTITUDE **2,200 m a.s.l.**	FLOCK **10,000 birds**	MIXER **1,000 kg batch**

A farm at that height is common in central Mexico and the Andes. As you will see in Block 3, altitude changes what is worth formulating, and the cost too.

0.6 The path in eight steps

The first time takes about 20 minutes. After that, with your data saved, updating prices and reformulating takes under two.

STEP 1 — 1 min
Genetics
Line and sex of your birds

STEP 2 — 2 min
Phase
Phase and expected performance

STEP 3 — 2 min
Conditions
Altitude, heat, flock, mixer

STEP 4 — 3 min
Ingredients
The ones you can buy

STEP 5 — 5 min
Prices
Your cost and the limits

STEP 6 — 3 min
Formulation
The diet and mixing sheet

STEP 7 — 2 min
Program
All phases, cost per bird

STEP 8 — 1 min
Report
Print, export and save

BLOCK 1 · GENETICS

1 Genetic line and sex

The line sets how fast your bird grows and, with it, how concentrated its feed must be.

1.1 Choose your genetic line

Each breeding company publishes how much energy, protein and amino acids its bird needs in each phase. RatioAvis has those tables built in: just tap the card for the line you raise.

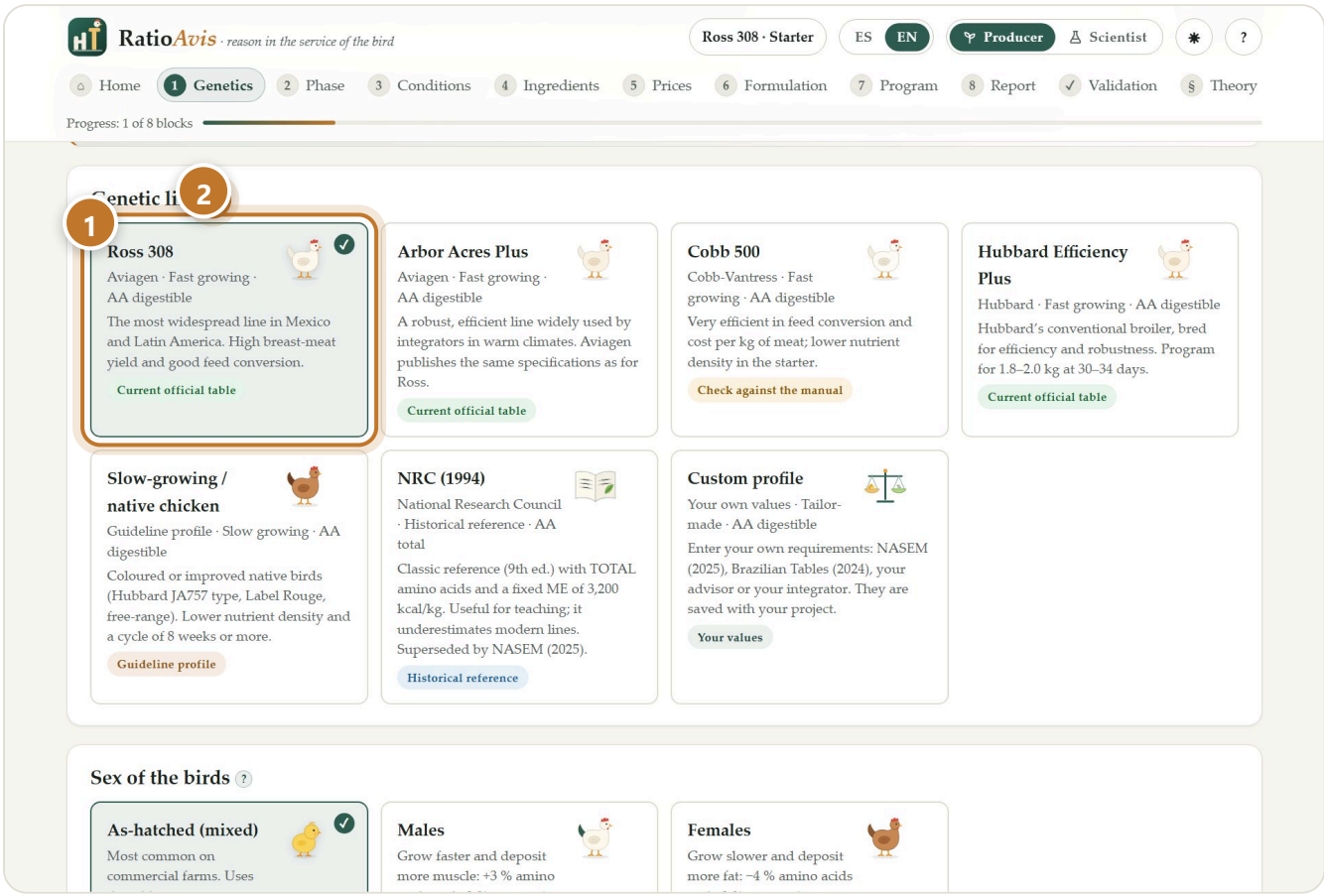


FIGURE 1.1 **Block 1.** The selected card shows a tick. The coloured label tells you how reliable the table is.

1. **Ross 308**, the line in our case.
2. The **?** explains what a genetic line is and how to choose it.

→ **DECISION RULE · WHAT DOES THE LABEL MEAN?**

Current official table	Use it as is. It is the one published by the breeder.
Check against the manual	Compare the Block 2 values with the current manual from your chick supplier.
Guideline profile	It is a starting point. Adjust it with your advisor and your records.
Historical reference	NRC (1994). Useful for teaching; it falls short for today's birds.
Your values	Custom profile: you enter your own requirements in Block 2.

✓ **TIP**

Raising native, free-range or coloured birds? Choose **Slow-growing / native chicken**. These birds eat a less concentrated feed and have a cycle of 8 weeks or more.

1.2 Set the sex of the birds

Below the lines is the **Sex of the birds** card. Almost every commercial farm raises **as-hatched (mixed)**, and that is the default.

OPTION	WHAT IT DOES TO THE TABLE
As-hatched (mixed)	Uses it as is.
Males	+3 % amino acids and +2 % protein: they grow faster and lay down more muscle.
Females	−4 % amino acids and −2 % protein: they grow slower and lay down more fat.

◆ **ON THE FARM**

Our chicks are **Ross 308** and arrive unsexed: we tap **Ross 308** and leave **As-hatched**. The app notes below where the requirements come from (Aviagen, 2022) and we move on with Next →.

BLOCK 2 · PHASE

2 Growth phase and requirements

As the bird grows it needs less protein and more energy per kilo of feed. That is why one diet is formulated per phase.

2.1 Choose the phase

Each line splits the bird's life into three or four phases. Tap the one you will formulate; each card shows its days, energy and lysine.

ROSS 308	DAYS	ENERGY (KCAL/KG)	PROTEIN	DIG. LYSINE
Starter	0–10	2,975	23.0 %	1.32 %
Grower	11–24	3,050	21.5 %	1.18 %
Finisher 1	25–39	3,100	19.5 %	1.08 %
Finisher 2	40 – market	3,125	18.0 %	1.02 %

2.2 Enter the expected performance

The **Expected performance in the phase** card asks how much feed each bird eats and how much weight it gains in that phase. The app fills in the table values, but **yours are better**: use the average of your last flocks. From them it computes the **feed conversion ratio** and, later, the cost per bird.

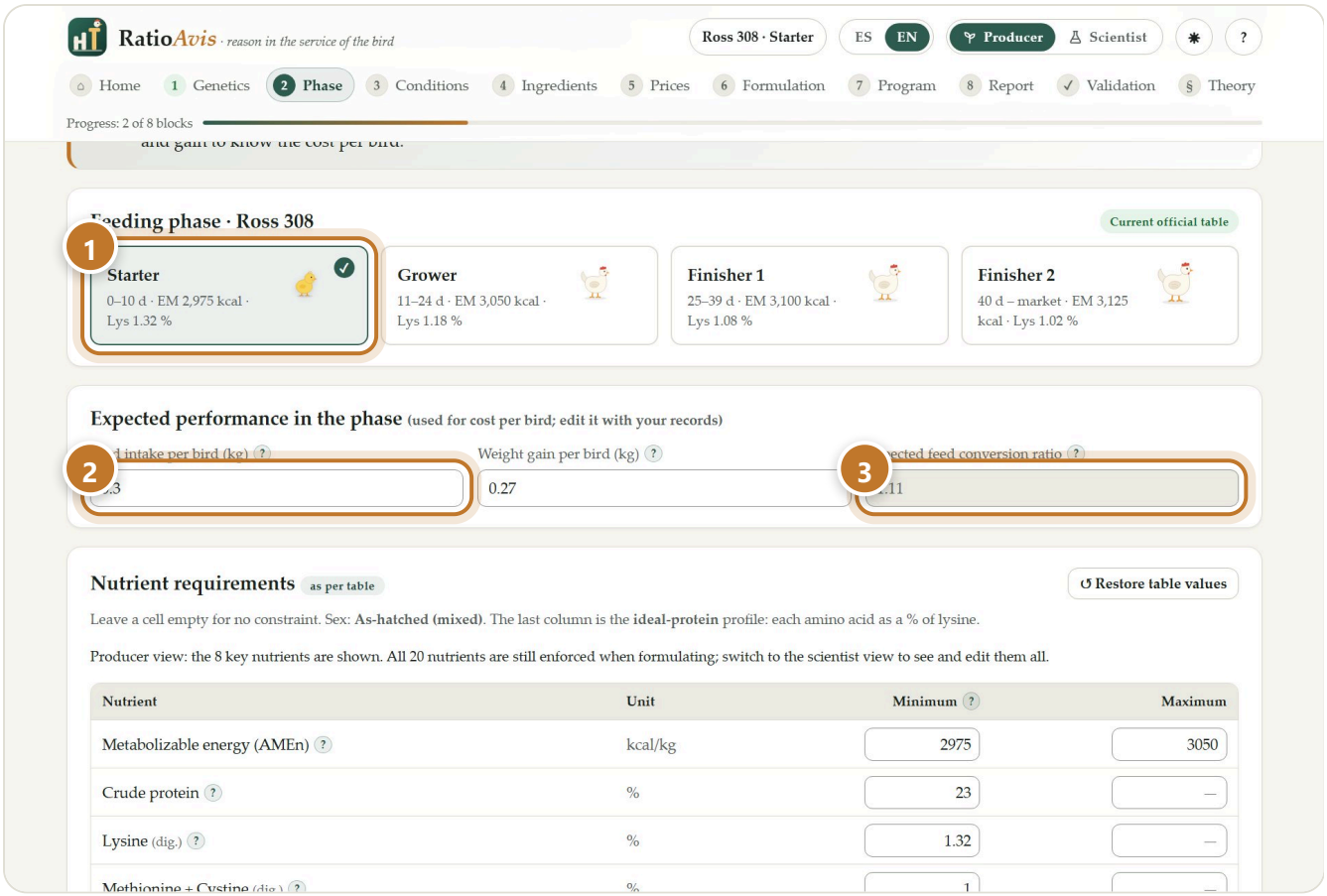


FIGURE 2.1 **Block 2.** Phase, performance and, further down, the requirements table.

1. **Starter** selected: days 0 to 10.
2. **Intake per bird** in the phase: 0.30 kg. Replace it with your records.
3. **FCR**: the app computes it (0.30 ÷ 0.27 = 1.11).

2.3 Requirements: look, but don't touch

Further down is the **Nutrient requirements** table, with each nutrient's minimum and maximum. In the Producer view you see the **8 key nutrients**: energy, protein, lysine, methionine + cystine, threonine, calcium, available phosphorus and sodium. The other 12 are still enforced when formulating, even if you do not see them.

> **!** **CAUTION**
>
> Change a requirement only if your nutritionist tells you to. If you edit one, the **edited** label appears; to go back to the original table use **↺ Restore table values**. An empty cell means "no limit".

BLOCK 3 · CONDITIONS

3 Production conditions

Altitude, heat and your market change what is worth formulating. This block recommends; you decide.

3.1 Your farm's data

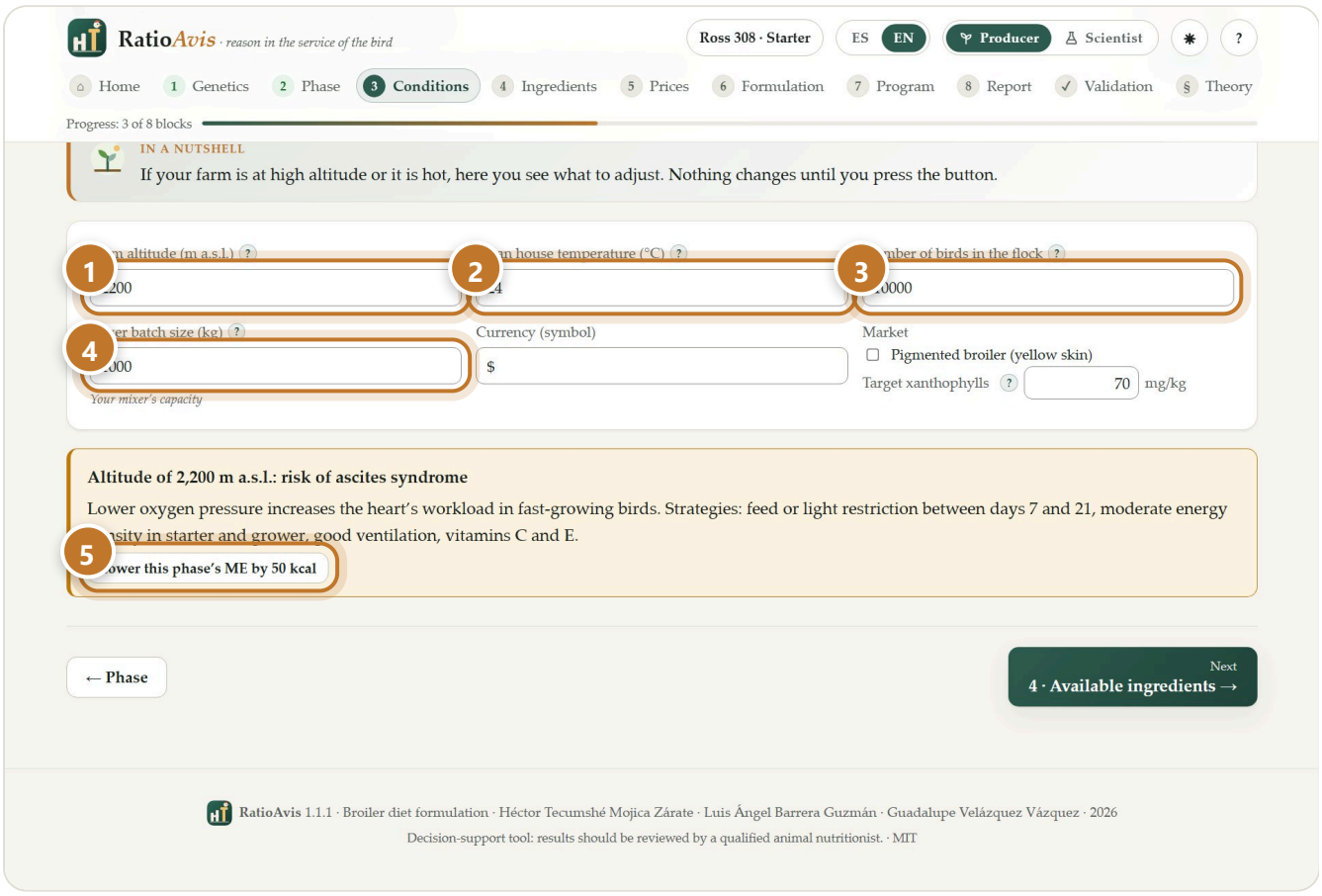


FIGURE 3.1 **Block 3 with the case data.** At 2,200 m the app detects an ascites risk and proposes an adjustment.

1. **Altitude** in metres above sea level.
2. **Mean temperature** inside the house, not in town.
3. **Number of birds** in the flock: gives the total feed needed.
4. **Batch**: what your mixer holds in one load.
5. **Adjustment button**: applies the recommendation to the Block 2 requirements.

Also type the **currency** (the symbol shown on costs) and, if you sell yellow-skinned birds, tick **Pigmented broiler**.

3.2 What the app recommends, and when

→ **DECISION RULE**

IF…	THE APP PROPOSES…
● altitude ≥ 1,500 m	**Lower this phase's ME by 50 kcal**, so as not to strain the bird's heart (ascites).
● temperature ≥ 30 °C	**Electrolyte balance of 230–300 mEq/kg**: birds eat less in the heat.
● temperature ≤ 18 °C	Only a note: check the heating; birds spend energy keeping warm.
● pigmented broiler	**Xanthophylls** (≈ 70 mg/kg) in the last phases, and choose the pigment in Block 4.
● none of the above	"No special adjustments".

Nothing changes until **you** press the recommendation's button.

? **WHY LOWER ENERGY AT ALTITUDE?**

Above 1,500 m there is less oxygen. A fast-growing bird on a very energy-dense feed grows faster than its heart and lungs can sustain, and ascites syndrome appears, a major cause of mortality. Moderating energy in the early phases slows that early growth slightly.

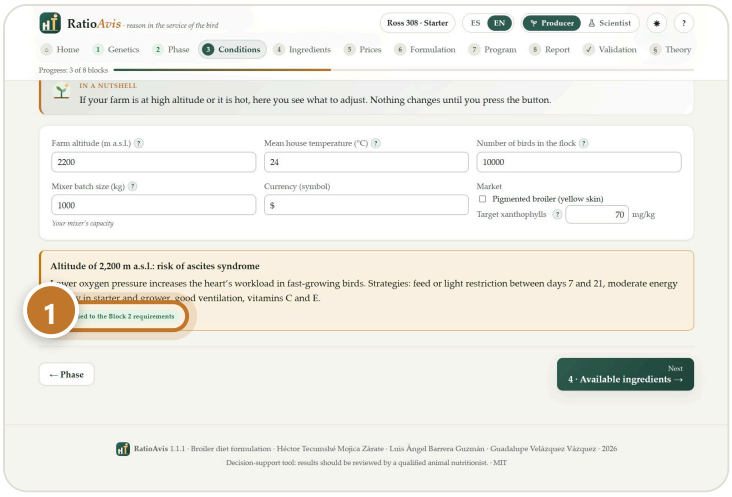


FIGURE 3.2 **Adjustment applied.** The button turns into a green confirmation.

ON THE FARM

We type **2200** altitude, **24** °C, **10000** birds and a **1000** kg batch. The app warns of ascites risk and we press **Lower this phase's ME by 50 kcal**.

The starter's minimum energy drops from 2,975 to **2,925 kcal/kg**. In Block 6 we will see how much money this saves.

BLOCK 4 · INGREDIENTS

4 Available ingredients

The app can only use what you tick. Tick what you can actually get.

4.1 Tick what you can buy

The library holds **29 ingredients** grouped by type: grains, protein meals, oils, minerals, amino acids and additives. The 14 of a typical corn–soy diet are ticked by default.

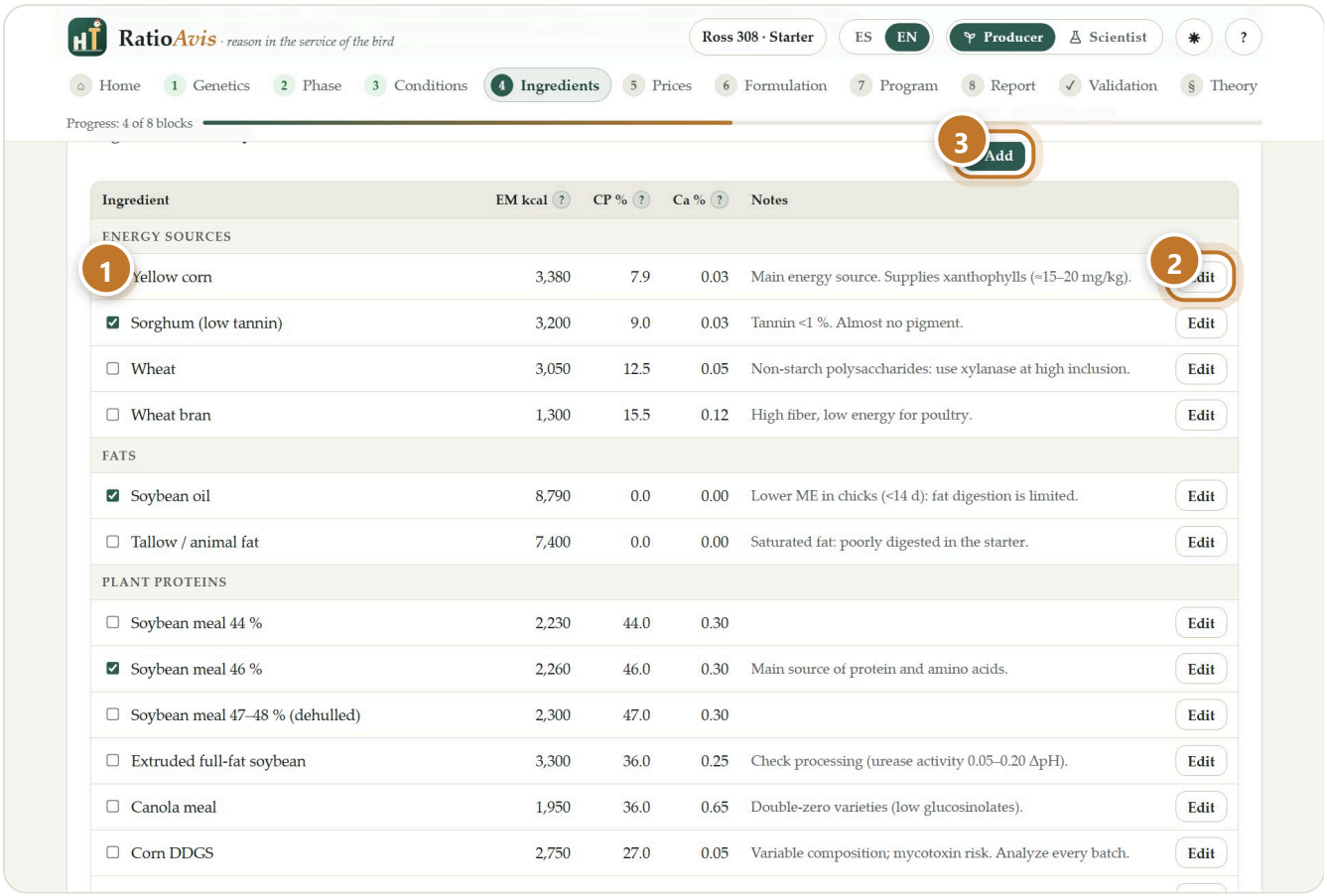


FIGURE 4.1 **Block 4.** Each row shows the ingredient's energy, protein, calcium and a practical note.

1. **Checkbox**: ticked, the app may use it; unticked, it may not.
2. **Edit**: change its composition with your lab analysis.
3. **Add**: create an ingredient that is not on the list.

→ **DECISION RULE**

- **If** you cannot get an ingredient for the whole cycle, **untick it**: a formula you cannot make is useless.
- **If** you have a cheap ingredient that is not listed (bran, local fishmeal…), **add it** with its analysis.
- **Always** keep the vitamin–mineral premix, salt and a calcium and phosphorus source ticked.

! **CAUTION · VALUES COME FROM TABLES**

Composition comes from recognised tables (Rostagno, FEDNA, NRC) and is expressed "as fed", at about 88–90 % dry matter. A wet corn or a lower-protein soybean meal changes the result. If you have a recent analysis, enter it with **Edit**; to go back to the original value use **↺ Table values**.

◆ **ON THE FARM**

Our supplier stocks corn, sorghum, soybean meal and the usual minerals and amino acids: we keep the 14 ingredients already ticked.

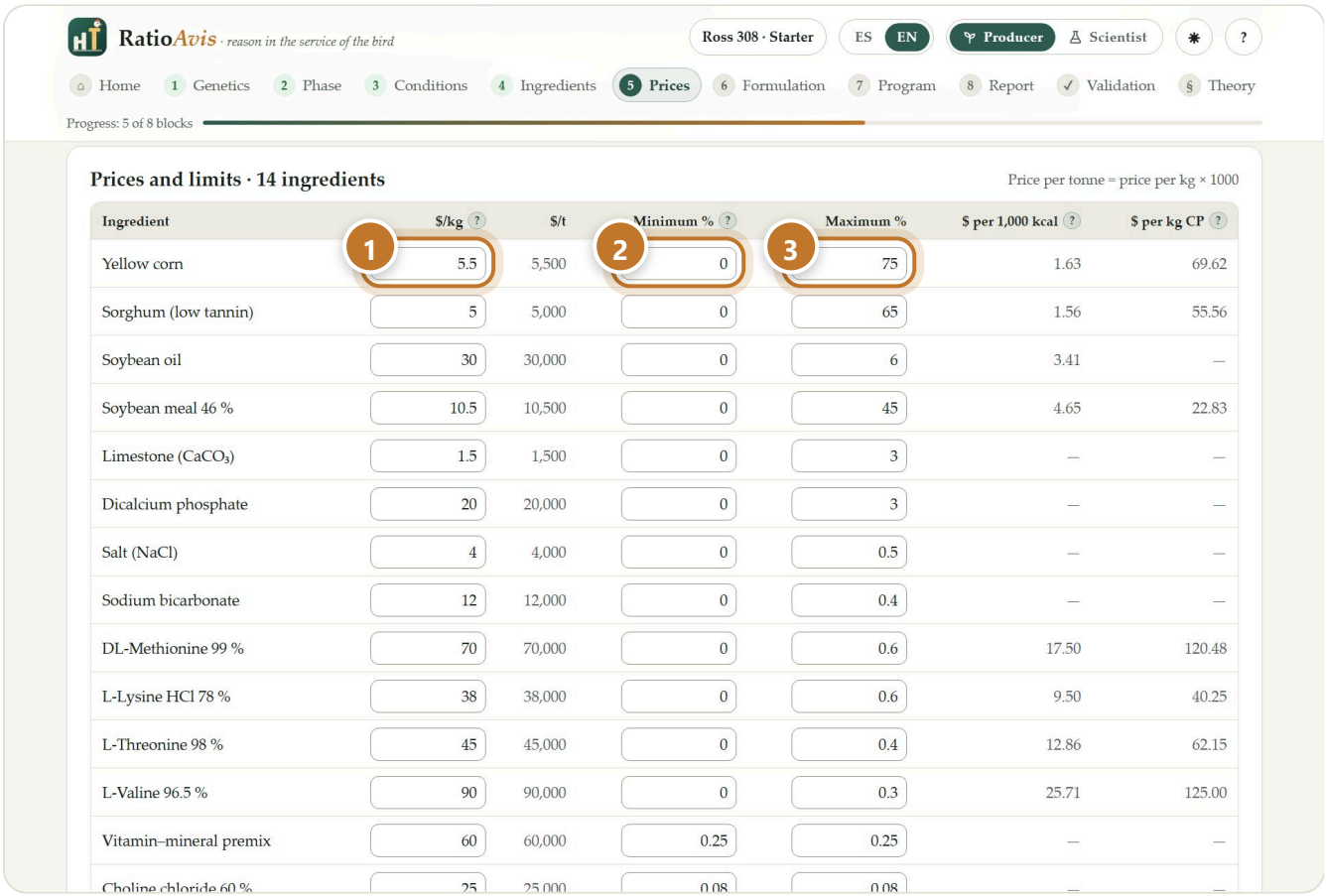
BLOCK 5 · PRICES

5 Local prices and inclusion limits

The result is only as good as your prices. This is the block most worth getting right.

5.1 Enter what you pay

For each ticked ingredient enter the **price per kilo delivered to your farm**, freight and handling included. The app works out the price per tonne.



Ingredient	$/kg	$/t	Minimum %	Maximum %	$ per 1,000 kcal	$ per kg CP
Yellow corn	5.5	5,500	0	75	1.63	69.62
Sorghum (low tannin)	5	5,000	0	65	1.56	55.56
Soybean oil	30	30,000	0	6	3.41	—
Soybean meal 46 %	10.5	10,500	0	45	4.65	22.83
Limestone ($CaCO_3$)	1.5	1,500	0	3	—	—
Dicalcium phosphate	20	20,000	0	3	—	—
Salt (NaCl)	4	4,000	0	0.5	—	—
Sodium bicarbonate	12	12,000	0	0.4	—	—
DL-Methionine 99 %	70	70,000	0	0.6	17.50	120.48
L-Lysine HCl 78 %	38	38,000	0	0.6	9.50	40.25
L-Threonine 98 %	45	45,000	0	0.4	12.86	62.15
L-Valine 96.5 %	90	90,000	0	0.3	25.71	125.00
Vitamin–mineral premix	60	60,000	0.25	0.25	—	—
Choline chloride 60 %	25	25,000	0.08	0.08	—	—

FIGURE 5.1 **Block 5.** One row per ticked ingredient. The last two columns compare how cheap energy and protein are from each.

1. **Price per kg** delivered to the farm.
2. **Minimum %**: the least the diet must contain.
3. **Maximum %**: the most it may contain.

> ! **CAUTION · THE PRELOADED PRICES ARE EXAMPLES**
>
> The preloaded prices are only for trying the app. Before mixing, **replace them with this week's quotes**: a small change in the price of corn changes the formula.

5.2 Inclusion limits

Limits prevent cheap but impractical or risky formulas: too much oil that does not mix well, too much sorghum that does not pigment, too much salt. The app comes with sensible limits; change them only if you have a reason.

> ✓ **TIP · FIXED INCLUSION**
>
> To make an ingredient go in at an exact amount, enter the **same value as minimum and maximum**. That is how the vitamin–mineral premix comes: `0.25` and `0.25`, as its label usually indicates.

5.3 Reading the last two columns

$ per 1,000 kcal and **$ per kg CP** tell you what the energy and protein from each ingredient cost you. Between two grains, the one with the lower energy figure is the cheaper energy source. In the example, sorghum supplies energy at 1.56 and corn at 1.63. Even so, the final formula depends on all nutrients at once, which is why you formulate.

BLOCK 6 · FORMULATION

6 Least-cost formulation

One button, and you get the diet, its cost per bird and a mixing sheet ready to print.

6.1 Press Formulate

At the top of the block you see a summary of your inputs: line, sex, phase, number of ingredients and, if you changed something, the **edited requirements** label. Check it and press **⚙ Formulate**.

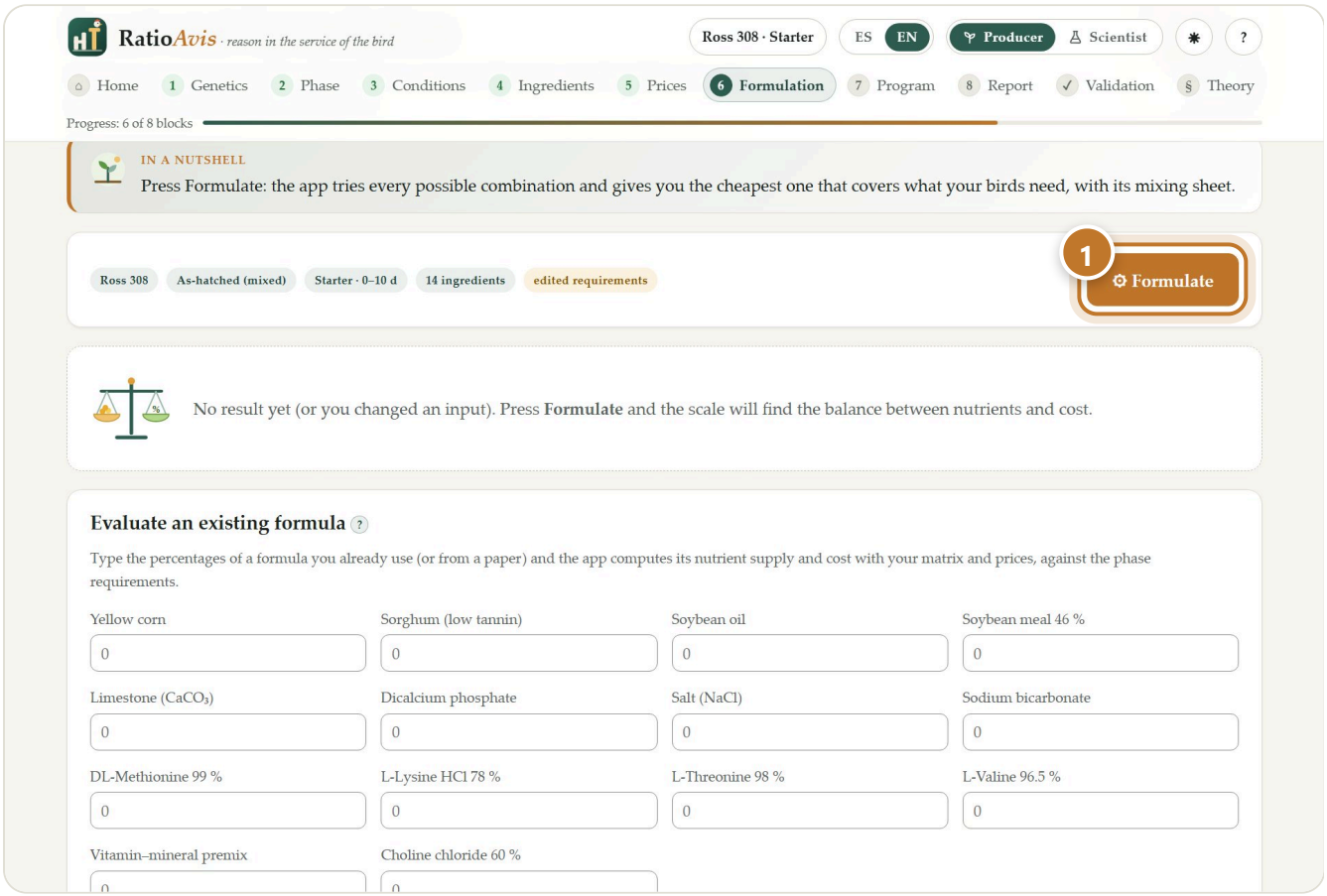


FIGURE 6.1 **Before formulating.** The summary confirms the app will work with your inputs.

6.2 The result in four numbers

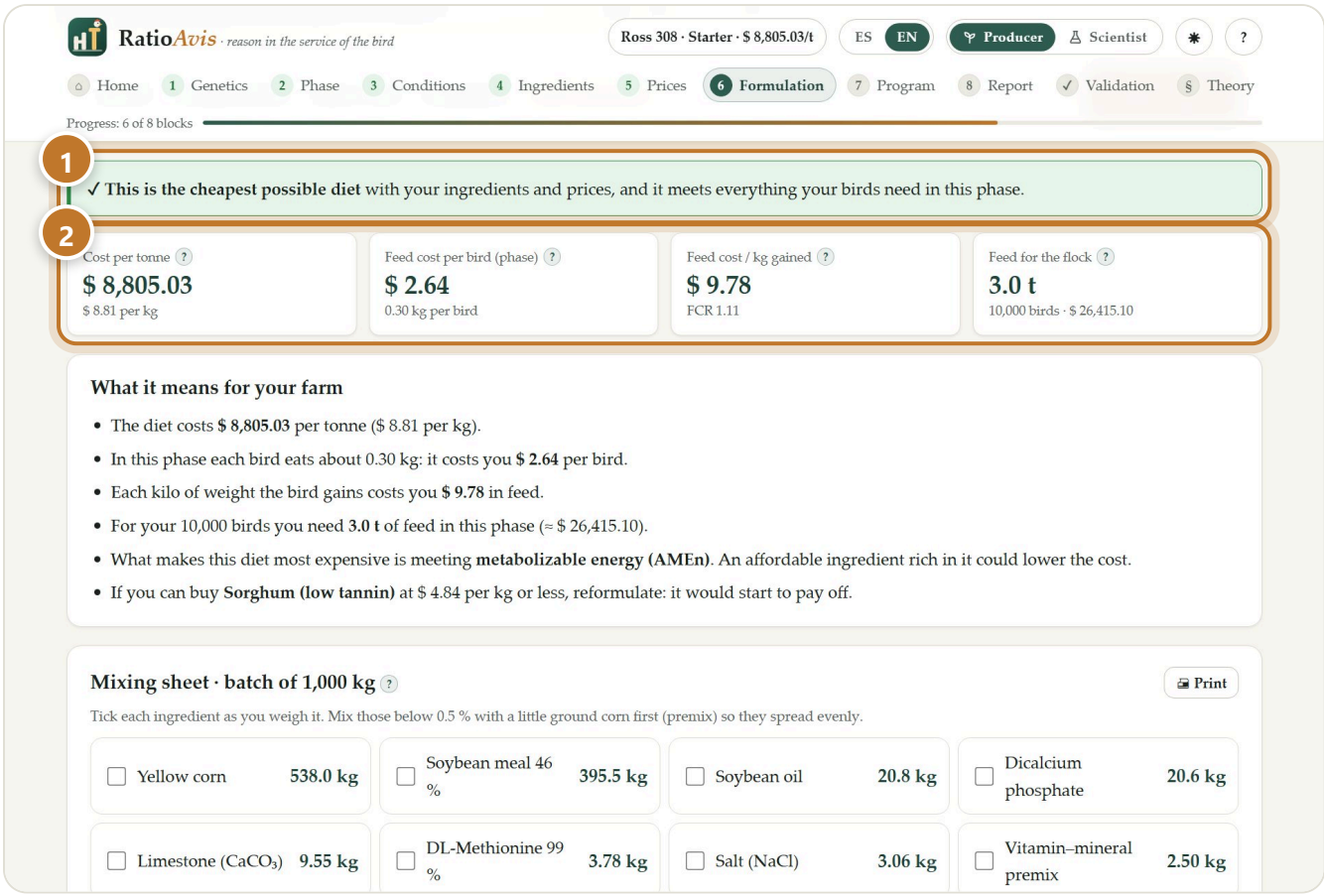


FIGURE 6.2 **Result of the case.** The green band confirms it is the cheapest possible diet and that it meets every requirement.

1 **Confirmation.** If you see a red box instead of green, go to section 6.7.

2 **The four cards** explained below.

COST PER TONNE	FEED PER BIRD	PER KG GAINED	FOR THE FLOCK
$ 8,805	$ 2.64	$ 9.78	3.0 t
$ 8.81 per kg	0.30 kg in the phase	FCR 1.11	$ 26,415

CARD	WHAT IT TELLS YOU
Cost per tonne	What it costs to make 1,000 kg of this diet. It is the figure to compare against commercial feed.
Feed cost per bird (phase)	Cost per kg × what each bird eats in the phase.
Feed cost / kg gained	How much feed you pay for each kilo the bird gains. The best figure for comparing diets.
Feed for the flock	Tonnes you need for your birds in this phase, and their cost. Useful for purchasing.

6.3 “What it means for your farm”

Below the cards, the app turns the result into sentences. The last two are the most valuable: they tell you **what drives your diet’s cost** and **which ingredient could cut it** if you find a better price.

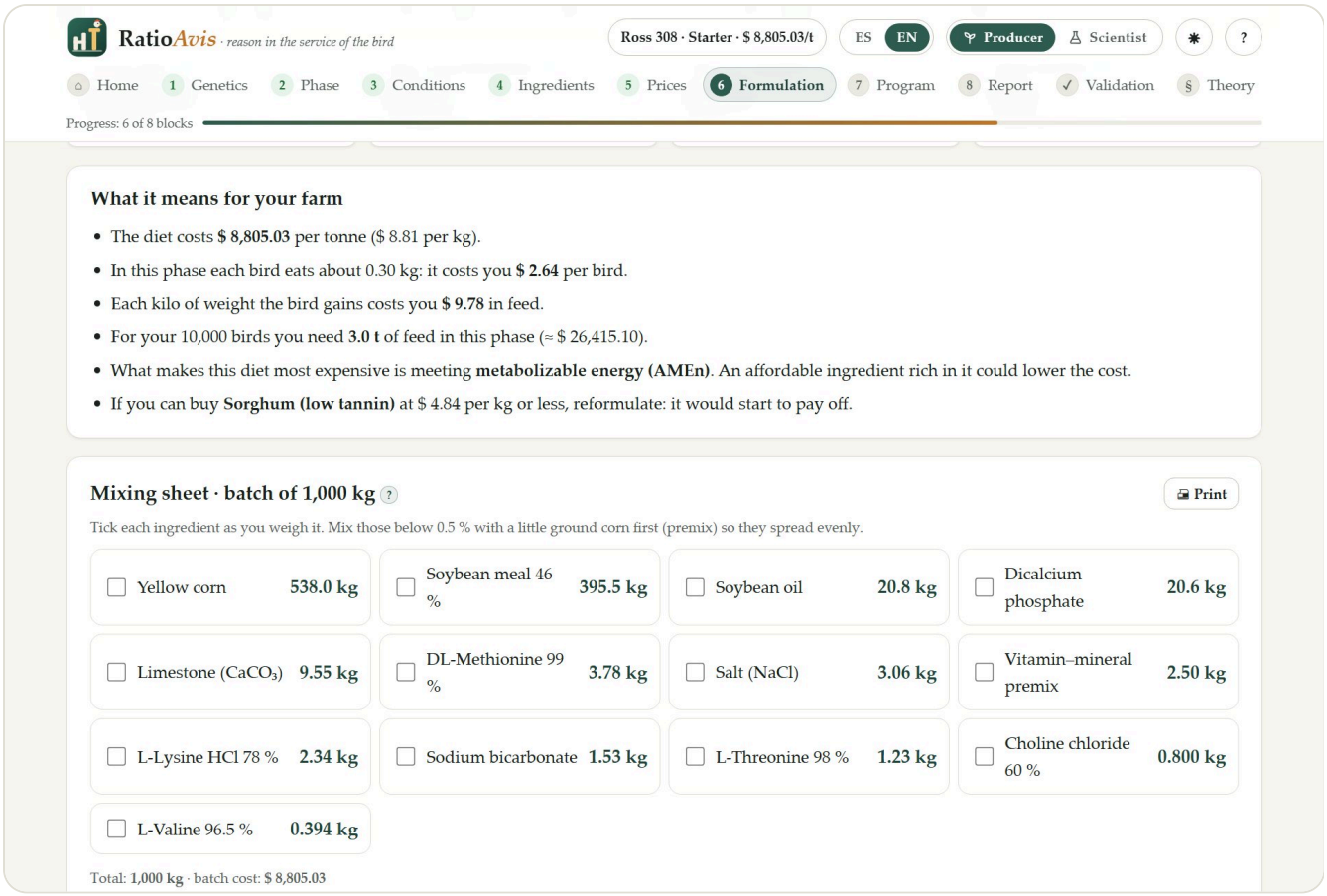


FIGURE 6.3 **Plain-language reading.** In the case, energy is the main cost driver, and sorghum would start to pay off at $4.84 per kg or less.

6.4 The mixing sheet

This is what you take to the mill: the exact kilos of each ingredient for **one batch** of your mixer, largest first, with a box to tick as you weigh. The Print button puts it on paper. On a phone it shows as a single column, handy to keep next to the scale.

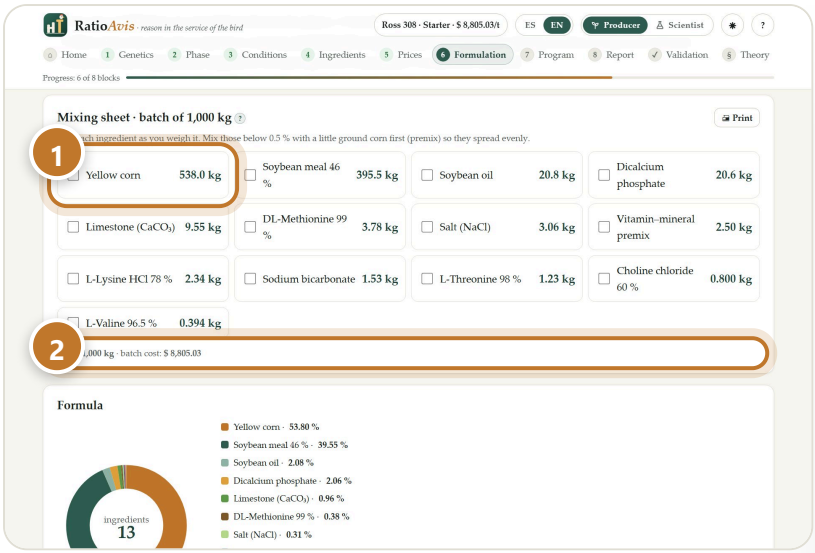


FIGURE 6.4 **The case's mixing sheet.** 13 ingredients, 1,000 kg, $8,805 per batch.

1. **Ingredient and kilos**; tick the box once weighed.
2. **Batch total** and what it costs.

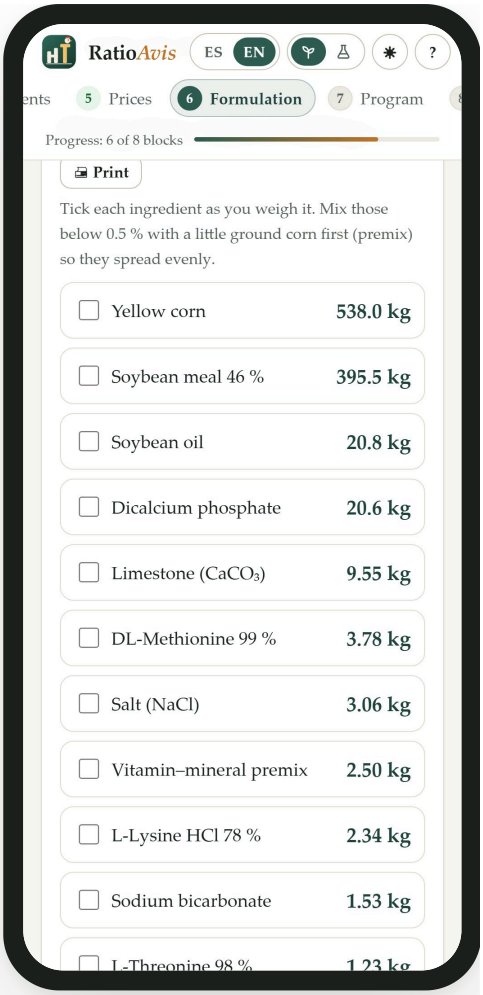


FIGURE 6.5 **On a phone.**

> **!** **CAUTION · INGREDIENTS BELOW 0.5 %**
>
> Methionine, lysine, threonine, valine, choline, salt, bicarbonate and premix go in at just a few kilos per tonne. Tipped straight into the mixer they do not spread evenly, and some birds get too much and others too little. **Blend them first with about 20 kg of ground corn** (a "premix") and add that premix to the batch.

◆ **ON THE FARM · FROM SHEET TO MIXER**

1. Weigh the 8 small ingredients (≈ 15.6 kg in total) and blend them with 20 kg of the batch's corn.
2. Load half of the remaining corn, then the soybean meal, phosphate and limestone.
3. Add the premix from step 1 and the rest of the corn.
4. Last, the oil (20.8 kg), slowly and with the mixer running.
5. Mix for the time your mixer's manufacturer recommends.

This is a general, practical loading order. If your mill already has a validated procedure, follow it.

6.5 What the altitude adjustment was worth

We formulated the same case twice: with the original table and with the −50 kcal adjustment from Block 3. Oil, which costs more than five times as much as corn, dropped from 3.04 % to 2.08 % of the formula:

ORIGINAL TABLE · 2,975 KCAL
$ 9,052 /t

→

ADJUSTED · 2,925 KCAL
$ 8,805 /t

That is **$247 less per tonne**; for the flock's 3 tonnes of starter, about **$740**. Besides saving money, it protects the birds from ascites. A good example of why Block 3 is worth filling in carefully.

6.6 Which ingredients are worth watching?

The **Unused ingredients · entry price** table tells you at what price an ingredient left out would start to pay off. Use it when negotiating with your supplier.

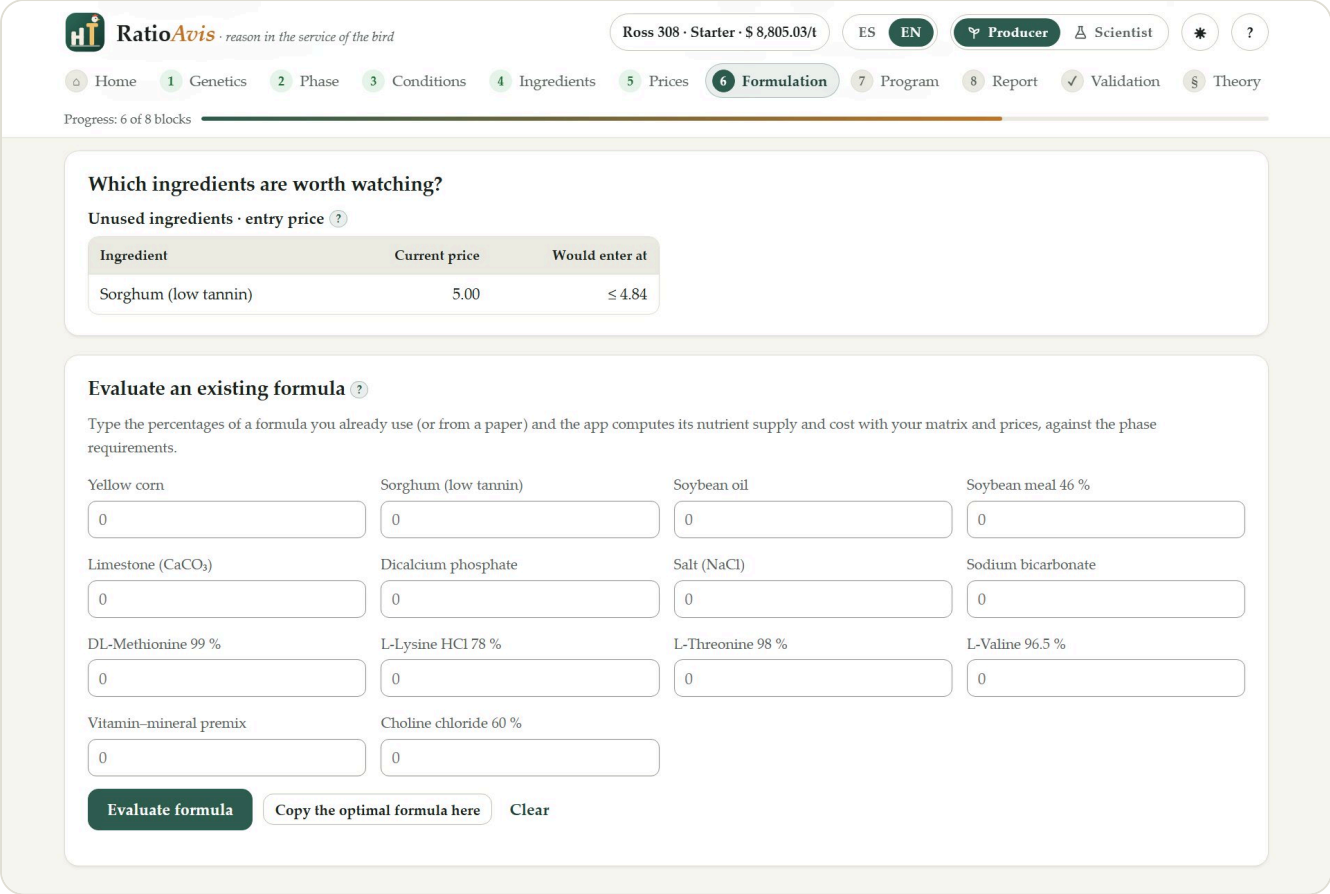


FIGURE 6.6 **Entry price.** Sorghum costs $5.00 and did not enter; at $4.84 or less it would. Below, **Evaluate an existing formula** computes the cost and nutrient supply of the formula you use today, to compare it with the optimum.

✓ **TIP · COMPARE AGAINST YOUR CURRENT FORMULA**

Type the percentages of the diet you use today into **Evaluate an existing formula** and press Evaluate formula. The **Difference vs. optimal formula** card shows how much extra you pay per tonne.

6.7 If “No solution” appears

It means that **no combination** of your ingredients meets everything at once. It is not an app error: it is information. RatioAvis tells you which constraint to relax so that there is a solution.

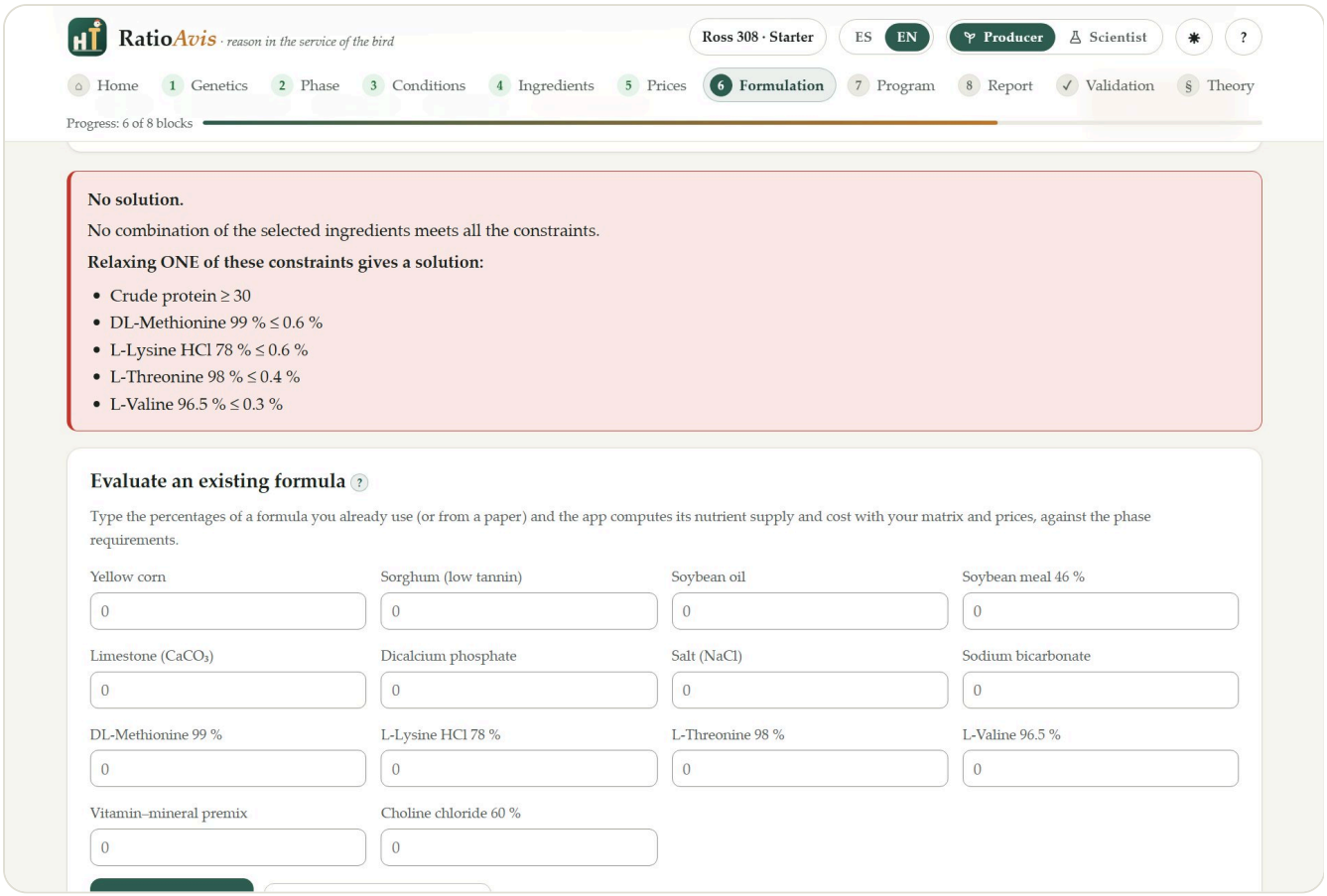


FIGURE 6.7 **Diagnosis.** Here an impossible 30 % protein was requested on purpose. The app shows that relaxing **one** of the listed constraints would be enough.

What do I do if there is no solution? Try in this order:

1 **Did you edit a requirement?**
Press *Restore table values* in Block 2 and formulate again.

2 **Does the app say "No selected ingredient supplies…"?**
Tick an ingredient in Block 4 that supplies it: phosphate for phosphorus, salt for sodium, synthetic amino acids.

3 **Did you set a very low maximum?**
Check the maximums in Block 5, especially oil and amino acids.

4 **Still no solution?**
Relax the constraint the app lists and check with your nutritionist.

BLOCK 7 · PROGRAM

7 Multiphase feeding program

Every phase at once: what it costs to feed each bird from start to finish.

7.1 The cost of the whole cycle

Press **⚙ Formulate all phases**. With the same ingredients and prices, the app formulates every phase of your line and adds up what each bird eats and costs over the whole cycle.

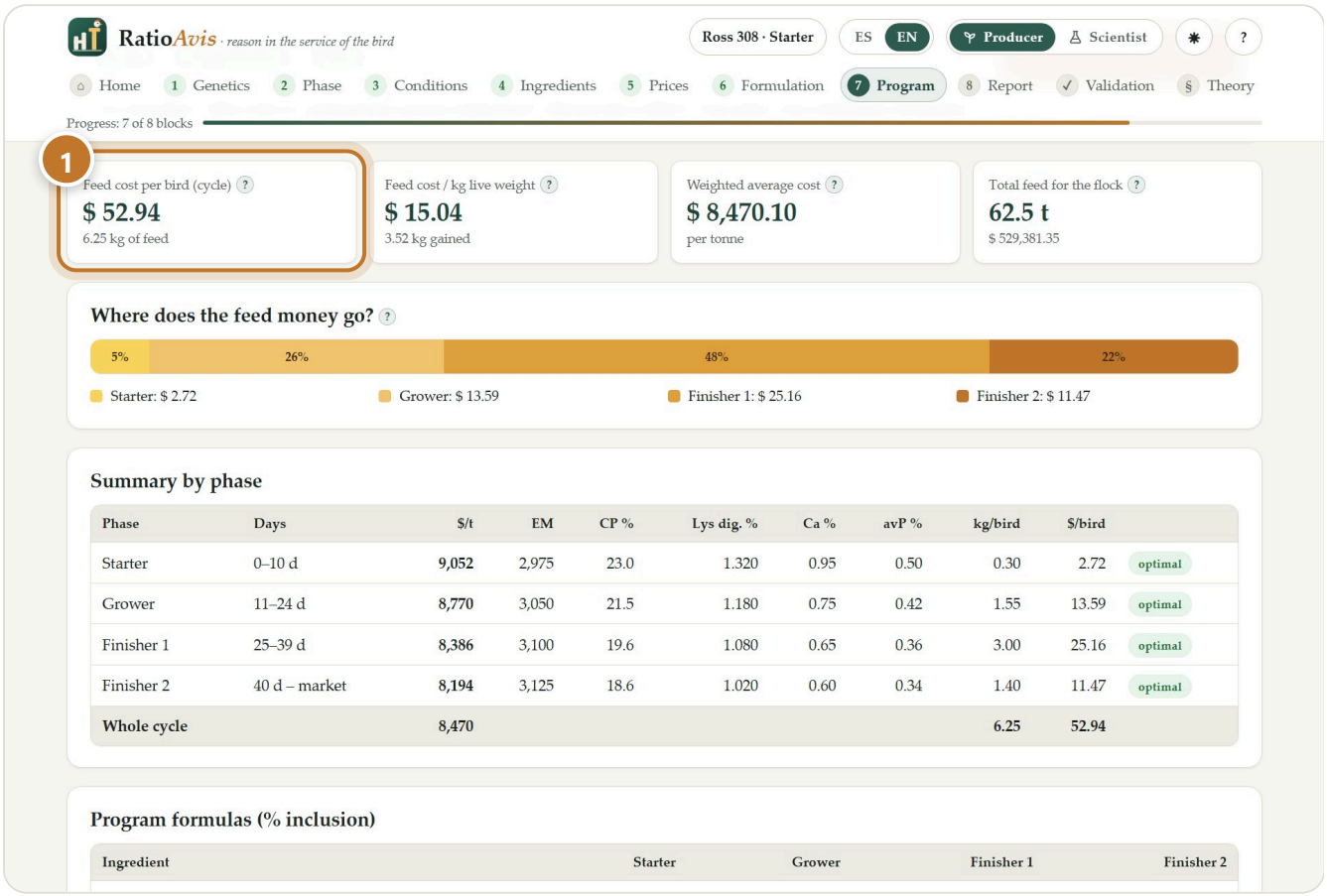


FIGURE 7.1 **The case's program.** Feeding each bird costs $52.94 over the cycle, and finisher 1 takes almost half the money.

1 **Feed cost per bird** over the cycle: the figure that matters most for your margin.

→ **DECISION RULE · WHERE TO LOOK FOR SAVINGS?**

Look at the **Where does the feed money go?** bar. **If** one phase takes the largest share (here finisher 1, at 48 %), **then** that is where negotiating prices or adjusting the formula pays off most. A small saving in that phase weighs more than a large one in the starter, which takes only 5 %.

! **CAUTION · THE PROGRAM USES EACH PHASE'S TABLE**

The program applies the sex, pigmentation and heat adjustments, but **not manual edits in Block 2** nor the altitude adjustment, which only apply to the phase you formulated in Block 6. That is why the starter shows here at $9,052 rather than $8,805. If your farm is at altitude, formulate each phase in Block 6 with its adjustment before mixing.

Further down, **Summary by phase** and **Program formulas** show each diet side by side; **⬇ Download program (CSV)** takes them to Excel.

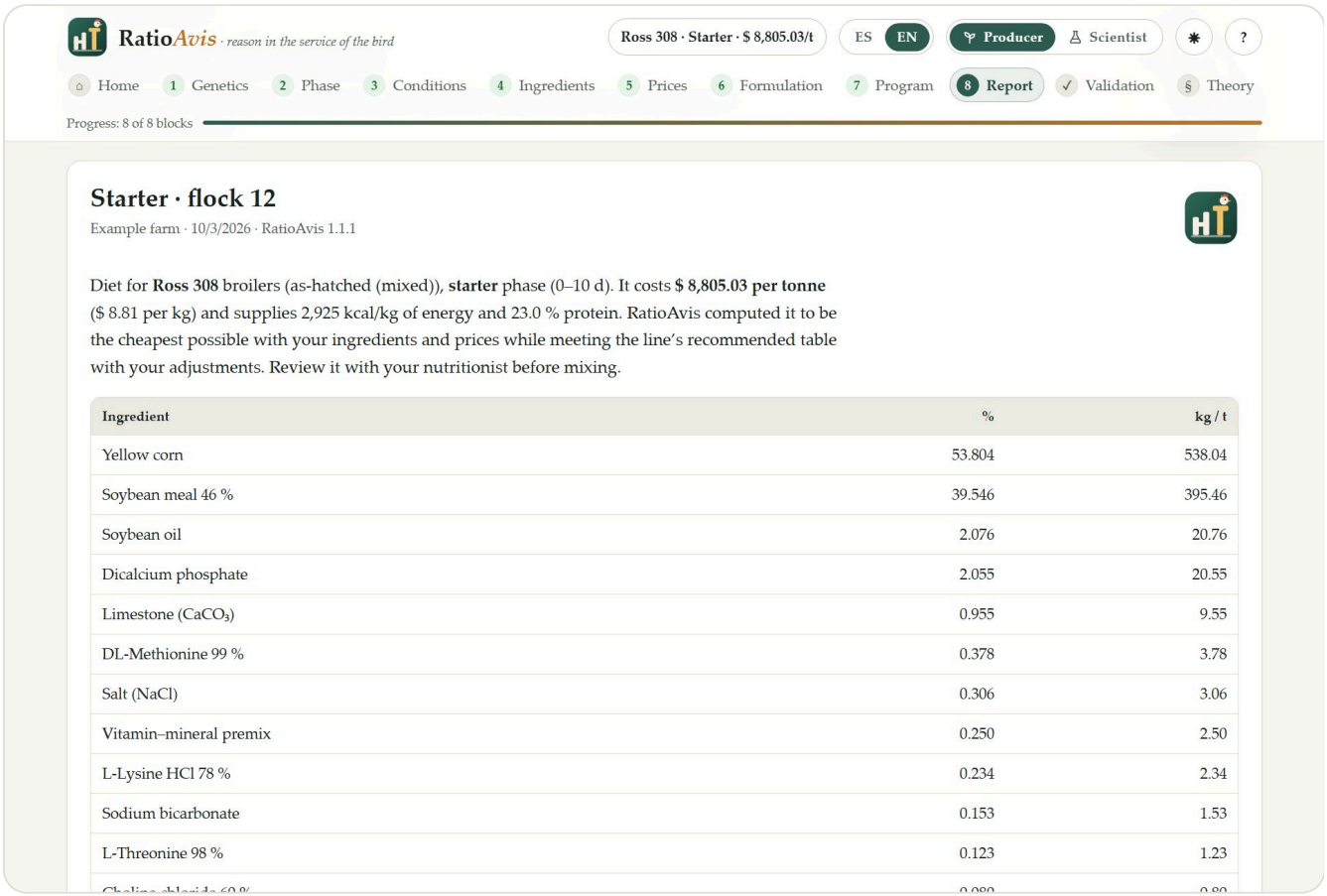
BLOCK 8 · REPORT

8 Report and export

Your diet on one page for your nutritionist, your tables in Excel and your project saved for next time.

8.1 The written report

The app writes a summary with your diet's and farm's names (entered on the Home screen under **Project details**), the date, the cost and the formula as percentages and kilos per tonne.



Ingredient	%	kg / t
Yellow corn	53.804	538.04
Soybean meal 46 %	39.546	395.46
Soybean oil	2.076	20.76
Dicalcium phosphate	2.055	20.55
Limestone ($CaCO_3$)	0.955	9.55
DL-Methionine 99 %	0.378	3.78
Salt (NaCl)	0.306	3.06
Vitamin–mineral premix	0.250	2.50
L-Lysine HCl 78 %	0.234	2.34
Sodium bicarbonate	0.153	1.53
L-Threonine 98 %	0.123	1.23

FIGURE 8.1 **The case's report.** Ready to print or save as PDF for your nutritionist.

8.2 Exporting and saving your project

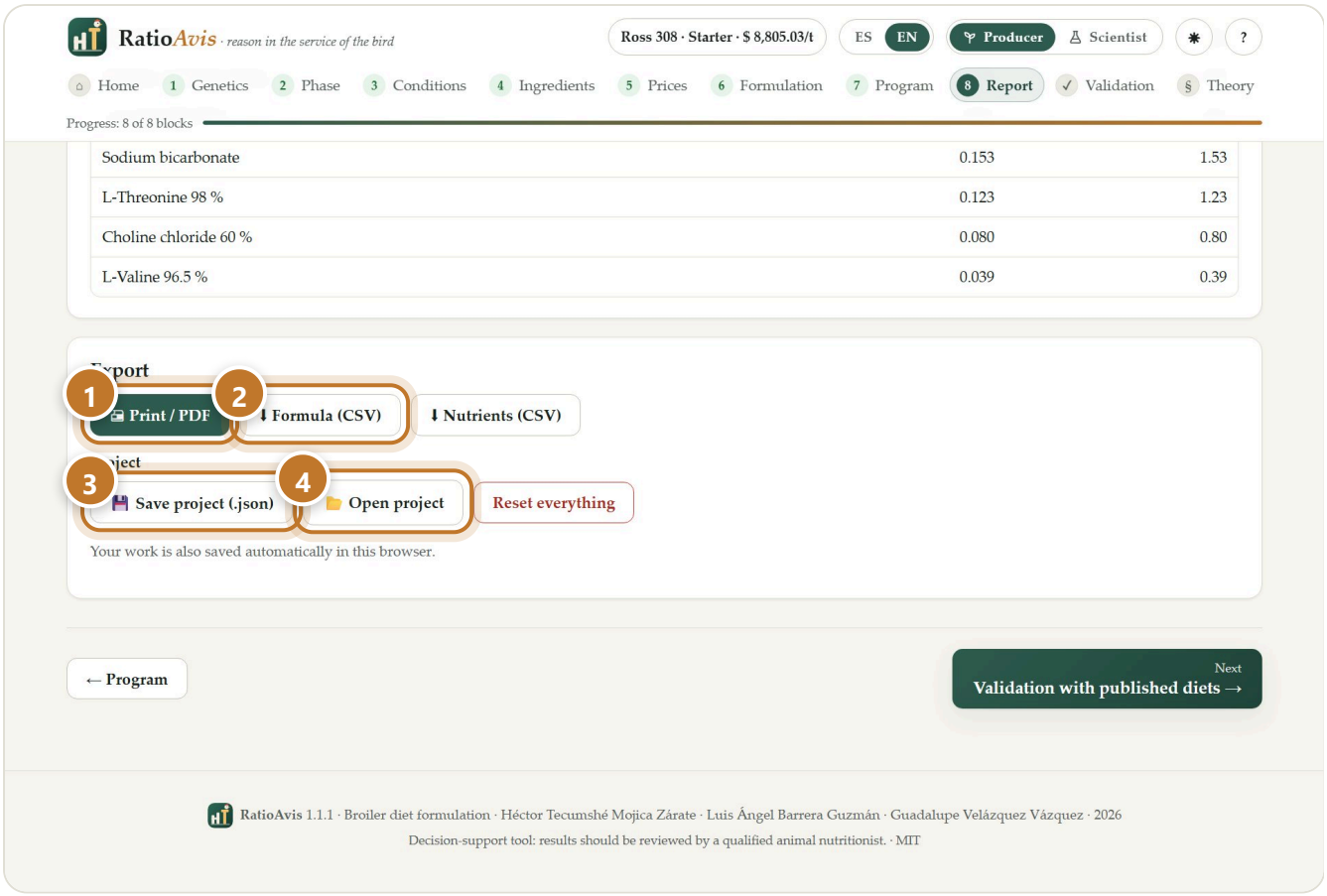


FIGURE 8.2 **The Export card.**

1. **Print / PDF**: the report on paper or as PDF.
2. **Formula (CSV)** and **Nutrients (CSV)**: tables for Excel.
3. **Save project (.json)**: all your work in one file.
4. **Open project**: reload that file on any computer.

✓ **TIP · YOUR WORK SAVES ITSELF, BUT…**

RatioAvis saves automatically in the browser you are using. If you switch computer or browser, or clear your history, it will not be there. So when you finish, use Report › Save project and keep the file with your flock records.

! **CAUTION · RESET EVERYTHING**

Reset everything erases your project and returns the app to its defaults. To prevent accidents it asks for confirmation: you must press it twice. Save your project first.

FROM PAPER TO MIXER

Before you mix

Seven checks that take five minutes and prevent a wrong batch of feed.

- [ ] The **prices** in Block 5 are this week's, with freight.
- [ ] The **phase** formulated matches the age of the birds that will eat it.
- [ ] Every ingredient in the formula is **in stock**, in the amount needed.
- [ ] The **batch** in Block 3 matches your mixer, and the scale is calibrated.
- [ ] If you analysed your ingredients, you entered the **lab values** in Block 4.
- [ ] Your **nutritionist** has reviewed the report. RatioAvis is a support tool, not a substitute.
- [ ] You saved the **project** (.json) with the flock date.

QUICK REFERENCE

Troubleshooting

The most common questions, with cause and fix.

I changed a price and the result disappeared.

The cost per bird shows “—”.

“No solution” appears.

I opened the app on another computer and my data are gone.

An ingredient I always use did not enter the formula.

The formula uses a lot of an ingredient that worries me.

Does switching language or view erase anything?

Can I use RatioAvis offline?

REFERENCE

Glossary

The app's words in plain farm language, with their Spanish name.

Ascites *Ascitis*	Fluid build-up in the abdomen from an overworked heart; common at altitude.
Batch *Bache*	What is mixed in one load of the mixer; for example, 1,000 kg.
Electrolyte balance *Balance electrolítico*	Sodium + potassium − chloride, in mEq/kg. In the heat, around 250 is advisable.
Feed conversion ratio (FCR) *Conversión alimenticia*	Kilos of feed per kilo of weight gained. The lower, the better.
Metabolizable energy (ME) *Energía metabolizable (EM)*	The feed energy the bird can actually use, in kcal/kg.
Phase *Fase*	Period of the bird's life with its own diet: starter, grower, finisher.
Inclusion limit *Límite de inclusión*	Minimum and maximum % an ingredient may have in the diet.
Digestible lysine *Lisina digestible*	The reference amino acid: the first one lacking in corn–soy diets.
Least cost *Mínimo costo*	The cheapest formula among all those that meet the requirements.
m a.s.l. *msnm*	Metres above sea level.
Entry price *Precio de entrada*	Price at which an ingredient left out would start to pay off.
Premix *Premezcla*	A pre-blend of ingredients under 0.5 % with a little corn, so they

	spread evenly.
Crude protein (CP) *Proteína cruda (PC)*	Total protein content of the feed, in %.
Delivered to farm *Puesto en granja*	Price including freight and unloading to your farm.
Requirement *Requerimiento*	Amount of a nutrient the bird needs per kilo of feed.
Xanthophylls *Xantofilas*	Yellow pigments (corn, gluten, marigold) that colour the skin.

PRINT IT AND PIN IT BY THE MIXER

RatioAvis on one page

hectortecumshe-ai.github.io/RatioAvis
Scan to open the app

1 Genetics
Tap your line (Ross 308, Cobb 500…) and sex. As-hatched if unsexed.

2 Phase
Pick the phase and enter your flocks' intake and gain.

3 Conditions
Altitude, temperature, birds and batch. Apply the adjustment if proposed.

4 Ingredients
Tick only what you can buy. Enter your analyses with Edit.

5 Prices
This week's price per kg delivered. Check the limits.

6 Formulation
⚙ Formulate. Print the mixing sheet. Premix anything below 0.5 %.

7 Program
⚙ Formulate all phases: cost per bird over the cycle.

8 Report
PDF for your nutritionist and Save project (.json).

IF "NO SOLUTION"

- Restore the Block 2 table.
- Tick the missing ingredient.
- Raise the maximum the app flags.

BEFORE MIXING

- This week's prices.
- Scale calibrated, batch correct.
- Report reviewed by your nutritionist.

RatioAvis · Producer Guide, 1st edition, October 2026. Covers RatioAvis 1.1.1. Every screenshot comes from the real app with the case data in section 0.5; prices are illustrative.

Decision-support tool: results should be reviewed by a qualified animal nutritionist. © 2026 Héctor Tecumshé Mojica-Zárate. The software is distributed under the MIT License.